



B28: SQL SuperHero Program : Session 2

1 message

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Delete/Truncate/DROP

1. delete is dml command and truncate is ddl command
2. delete can be rollback/commit but truncate can not be rollback or commit. it is autocommit
3. we can add condition like where clause in delete, but there is no condition in truncate
4. drop is also a ddl command and can not be rollback and no condition is drop also
5. when we drop a table, the whole table structure is gone but when we truncate only data is gone, table structure remain in database
6. Truncate is much faster than DELETE
7. Truncate will reset the HWM but delete will not.

--> delete --> to delete data from table

truncate --> to completely delete data from table , Reset the HWM(high water mark)

What is HWM

HWM is the last block of the database where data is written.

For example assume that you have a million row table that takes 30 seconds to read. After deleting 900,000 rows, a full scan on the table will still take 30 seconds. This is because the table high water mark is not re-set after delete operations. As such, there are no easy SQL scripts that will reveal the high water mark for an Oracle table, but you can assume that it is the last extent that was allocated to the table for estimation purposes.

High water mark is the maximum amount of database blocks used so far by a segment. This mark cannot be reset by delete operations. Does not matter , you delete and commit, HWM not reset.

The high water mark level is just a line separate the used blocks and free blocks.

- The blocks above the HWM level is free blocks, they are ready to use.
- The blocks below the HWM level is used blocks, they are already used.

For example, if you delete some huge records from the database, that data will delete but the blocks are not ready to used, because that blocks are still below HWM level, so delete command never reset the HWM level,

At the same time you truncate the data, then the data will delete and that used blocks will goto above the HWM level, now its ready to used. now they consider has free blocks.

truncate is faster and delete is taking much time.

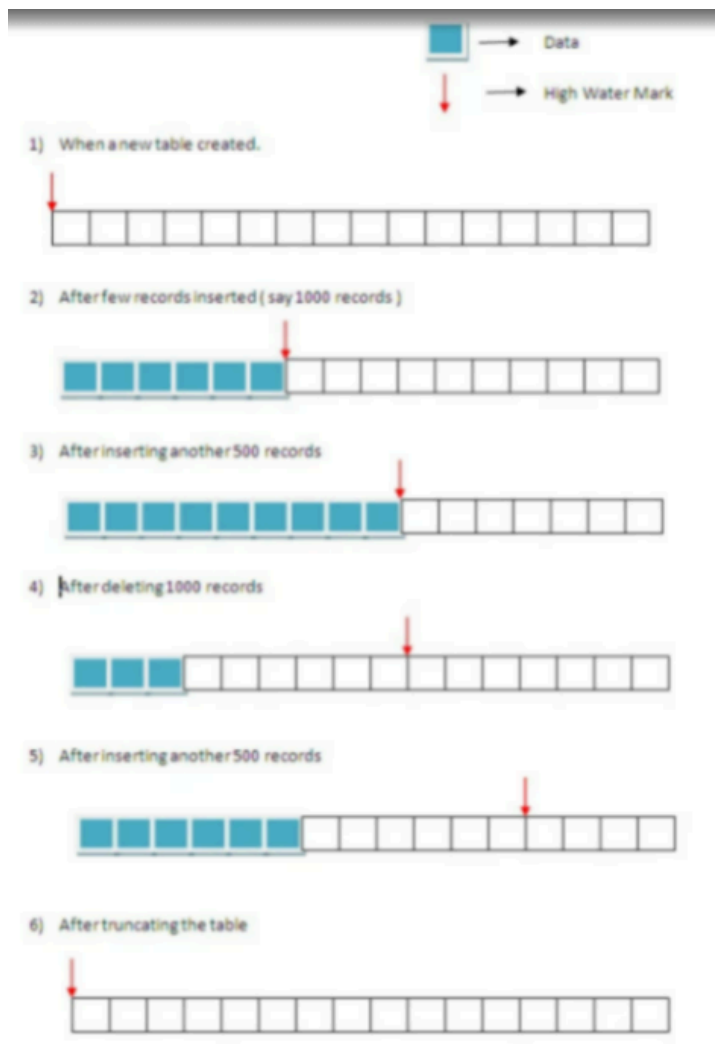
When we do delete, Oracle will store the data in redo log files, so that if the user wants it again, it reverts back from redo log.

but in case of truncate , no such things happen, it reset HWM and completely blank the table.

Why is rollback not working with truncate ?

Truncate is a DDL command. and rollback is only for DML.

3. **Can we do rollback after drop** --> NO. because drop is a ddl command. Rollback is not work for ddl only for dml



By default High Water Mark starts at the first block for new tables. When the table gets data via insert/update more blocks being used, High Water Mark moves forward. When data get deleted, High Water Mark remains the same even though blocks don't have any data. Delete has no impact on high water mark and Truncate reset the High Water Mark.

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Question: Table A has 10M records, I have to delete 5M., How we have to DO?

Answer:

1. we can not truncate here, because truncate will empty the whole table, which we do not want
 2. we can not fire the delete command, because deleting 5M rows from a table will definitely hung the database. it will stuck and don't know when come back
 3. Create table A_temp as select * from A where <condition> -- To fetch the required 5M records
 4. Now I have the required record which is needed for application in table A_temp
 5. truncate main table. truncate table A ; -- this will be done in a few seconds and now we have the application table empty and required data in temp table
 6. Either we can rename the temp table to the main table and drop the main table (This is an approach but not advisable because all the database objects like procedure, packages and function who are using that table, will become uncompiled and we have to compile again.)
- OR
- insert all records from temp table to main table

That is the one approach for deleting a high volume of data.

Note: Prefer to insert the data from temp table to main table, rather than drop the main table and rename the temp to main. Why? because if we drop the main table then all database objects (procedures, functions, packages etc) become invalid, we have extra work to compile them all.

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Super Key

A super key is a set of one or more than one key that can be used to identify a record uniquely in a table. Example: Primary key, Unique key, Alternate key are a subset of Super Keys. This key includes only those fields which include a unique value.

Candidate Key

A Candidate Key is a set of one or more fields/columns that can identify a record uniquely in a table. There can be multiple Candidate Keys in one table. Each Candidate Key can work as a Primary Key. A Candidate key is a subset of Super keys.

Alternate key

An Alternate key is a key that can work as a primary key. Basically, it is a candidate key that currently is not a primary key.

In other words, the alternate key is a column or collection of columns in a table that can uniquely identify each row in that table. There can be only one Primary key for a table. Therefore all the remaining Candidate keys are known as Alternate or Secondary keys

Surrogate key

A surrogate key is any column or set of columns that can be declared as the primary key instead of a “real” or natural key. The surrogate key is not derived from application data.

It is the sequential number outside of the database that is made available to the user and the application or it acts as an object that is present in the database but is not visible to the user or application.

We can say that, in case we do not have a natural primary key in a table, then we need to artificially create one in order to uniquely identify a row in the table, this key is called the surrogate key or synthetic primary key of the table. It turns out a surrogate key is very similar to a primary key in that it is a unique value for an object in a table. However, rather than being derived from actual data present in the table, it is a field generated by the object itself. A surrogate key is often a sequence number in a database such as a PostgreSQL serial column, Oracle sequence column, or MySQL auto_increment column.

[Remember the concept of medical term Surrogacy while answering this topic in interview]

KEYS CONTINUES IN NEXT SESSION.....